KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

# Sports Tournament Team Registration And Match Result System

Submitted by

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# DECLARATION

I hereby declare that the project entitled "**Sports Tournament Team Registration and Match Result System"** submitted in partial fulfillment of the requirements for the course **I B.SC IT-A** is a **record of my original work** carried out under the guidance of **DHILEEBAN A ,** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

**PAVANA SREE S**

# CERTIFICATE

This is to certify that the project entitled **"Sports Tournament Team Registration and Match Result System"** is a **bona fide work** carried out by [**PAVANA SREE S], [I B.SC IT -A]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

**Place:**

**Date:**

Submitted for the Viva-Voce Examination held on _____

**Guide:**
(Technical Trainer, KGCAS)

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**Head of the Department**
(Technical Head, KGCAS)

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**ABSTRACT**

The Sports Tournament Team Registration and Match Result System is a software application designed to simplify the management of sports events. This system allows teams to register easily by entering their details such as team name, players, and contact information. It also helps organizers schedule matches, record match results, and display rankings or winners efficiently. The system reduces manual work, minimizes errors, and saves time by automating the entire tournament process. Additionally, it provides a user-friendly interface for both administrators and participants, ensuring smooth coordination and better organization of sports tournaments.

# 1. INTRODUCTION

The Sports Tournament Team Registration and Match Result System is developed to provide an efficient and organized way of managing sports tournaments. In traditional methods, team registration and result handling are done manually, which can lead to errors, confusion, and time consumption. This system aims to digitize the entire process by allowing teams to register online, enabling organizers to schedule matches, and updating results instantly. It helps in maintaining accurate records, improving transparency, and ensuring smooth communication between participants and organizers. Overall, the system enhances the effectiveness and reliability of tournament management.

## 1.1 Objective

The main objective of the Sports Tournament Team Registration and Match Result System is to develop an efficient and user-friendly application for managing sports tournaments. The system aims to simplify team registration, automate match scheduling, and accurately record and display match results. It also focuses on reducing manual errors, saving time, and improving data management. Additionally, the system is designed to provide quick access to tournament information, ensure transparency, and enhance the overall organization and coordination of sports events.

## 1.2 Overview

The Sports Tournament Team Registration and Match Result System is a comprehensive application designed to manage all aspects of a sports tournament in a structured and efficient manner. The system provides features for team registration, player management, match scheduling, and result updating. It allows administrators to organize tournaments easily while enabling participants to view match details and results in real time. The system maintains a centralized database to store all information securely, ensuring accuracy and easy retrieval of data. By automating key processes, it improves efficiency, reduces paperwork, and enhances the overall experience of both organizers and players.

## 1.3 Features

The Sports Tournament Team Registration and Match Result System offers several important features to ensure smooth tournament management. It provides an easy team registration process where users can enter team and player details. The system includes match scheduling functionality to organize fixtures automatically or manually. It allows administrators to update match results and instantly display scores, rankings, and winners. Additionally, it supports data storage and retrieval, ensuring all tournament information is securely maintained. The system also includes a user-friendly interface, quick access to match details, and real-time updates, making it convenient for both organizers and participants.

# 2. PROGRAM ANALYSIS

The Sports Tournament Team Registration and Match Result System is analyzed to understand the requirements and functionalities needed for effective tournament management. The system identifies key user roles such as administrators and participants, and defines their interactions with the application. It examines the process of team registration, match scheduling, result updating, and data management. The analysis also considers system requirements like accuracy, reliability, and user-friendliness. Input data such as team and player details are processed to generate outputs like match schedules, scores, and rankings. By analyzing these components, the system ensures efficient performance, reduced errors, and smooth handling of tournament activities.

## 2.1 Problem Definition

The Sports Tournament Team Registration and Match Result System is a software application developed to manage and automate the key activities of a sports tournament. It is designed to handle team registration, store player information, schedule matches, and record match results in an organized manner. The program provides functionalities for administrators to control and monitor the tournament, while allowing users to access match details and results It uses a structured approach to process input data and generate accurate outputs such as fixtures, scores, and rankings. Overall, the program ensures efficient tournament management by reducing manual effort and improving data accuracy.

## 2.2 Project Overview

The Sports Tournament Team Registration and Match Result System is a software project developed to efficiently manage and organize sports tournaments. This system focuses on automating essential tasks such as team registration, player data management, match scheduling, and result processing. It provides a centralized platform where administrators can control tournament activities and participants can easily access match information and results. The project is designed using user-friendly interfaces and reliable data handling techniques to ensure accuracy and smooth operation. By replacing manual processes with a digital solution, the system improves efficiency, reduces errors, and enhances the overall experience of managing sports events.

## 2.3 Scopes and Limitations

### Scope:

The Sports Tournament Team Registration and Match Result System is designed to handle the core activities involved in managing sports tournaments. The scope of the system includes team registration, player information management, match scheduling, and result updating. It allows administrators to organize tournaments efficiently and enables users to view match details, scores, and rankings. The system can be used for different types of sports and small to medium-level tournaments. However, it is mainly focused on basic tournament management and may not include advanced features such as live streaming or complex analytics. Overall, the system aims to provide a reliable and easy-to-use solution for improving tournament organization and management.

### Limitations:

The Sports Tournament Team Registration and Match Result System has certain limitations that should be considered. The system is mainly designed for small to medium-scale tournaments and may not efficiently handle large-scale events with a high number of teams and matches. It may have limited features such as the absence of live match streaming, advanced statistical analysis, or real-time notifications. The system also depends on accurate data input by users, and any incorrect information may affect the results. Additionally, it may require internet access for full functionality and might have limited customization options based on specific tournament requirements.

# 3. SYSTEM DESIGN

The Sports Tournament Team Registration and Match Result System is designed using a structured and modular approach to ensure efficient performance and easy maintenance. The system consists of different modules such as team registration, player management, match scheduling, and result processing. A user-friendly interface is provided for both administrators and participants to interact with the system . The backend is designed to handle data storage using a database, ensuring secure and organized management of tournament information. The system follows a logical flow where input data is processed to generate outputs like match fixtures, scores, and rankings. Proper validation and error-handling mechanisms are included to maintain data accuracy and system reliability

## 3.1 Modules

The Sports Tournament Team Registration and Match Result System is divided into several modules to ensure smooth and organized functionality. The Team Registration Module allows users to enter and manage team and player details. The User Management Module handles login and access control for administrators and participants. The Match Scheduling Module is responsible for creating and organizing match fixtures. The Result Management Module enables updating and storing match results. The Ranking and Report Module generates rankings, scores, and tournament reports. Each module works together to provide an efficient and user-friendly system for managing sports tournaments.

### 1. Project  Management Module

The Sports Tournament Team Registration and Match Result System includes additional modules to enhance functionality and performance. The Admin Module allows administrators to manage the entire tournament, including teams, matches, and results. The Team Management Module is used to add, update, or delete team and player details. The Fixture Generation Module automatically creates match schedules based on the number of teams. The Score Update Module helps in entering and updating match scores efficiently. The Notification Module provides updates about match timings and results to users. The Database Management Module ensures secure storage, retrieval, and maintenance of all tournament data. These modules work together to provide a complete and efficient tournament management system.

**2. Inventory Management Module**

The Sports Tournament Team Registration and Match Result System also includes inventory management modules to handle sports equipment and related resources efficiently. The Equipment Management Module is used to add, update, and track sports items such as balkits, and other materials. The Stock Management Module monitors the availability and quantity of equipment to avoid shortages. The Issue and Return Module records the distribution and return of equipment to teams or players. The Maintenance Module keeps track of damaged or repairable items. The Report Module generates inventory reports, including usage and stock details. These modules help in maintaining proper control and efficient utilization of sports resources during tournaments.

**3. Additional Management Module**

This module manages all sales transactions in the system.

Allows users to select products and enter quantity

Calculates the total bill amount

Displays transaction details

Updates stock after purchase

It helps in performing fast and accurate billing operations.

**4. Interface Module**

The Sports Tournament Team Registration and Match Result System includes an Interface Module that provides a user-friendly environment for both administrators and participants to interact with the system. This module is designed with simple navigation, clear menus, and easy-to-use forms for tasks such as team registration, match scheduling, and result viewing. It ensures that users can access information quickly and perform actions without confusion. The interface supports proper input validation, error messages, and responsive design to improve usability. Overall, the Interface Module enhances user experience by making the system accessible, efficient, and easy to operate.

### 5. Data Handling Module

The Sports Tournament Team Registration and Match Result System includes a Data Handling Module that manages the storage, processing, and retrieval of all tournament-related information. This module is responsible for handling data such as team details, player information, match schedules, and results in a structured manner. It ensures that all data is stored securely in the database and can be accessed or updated whenever required. The module also performs data validation to maintain accuracy and consistency. Additionally, it supports quick data retrieval and report generation, making the system reliable and efficient for managing tournament information

### 4.1. Problem Analysis

The Sports Tournament Team Registration and Match Result System is developed after analyzing the problems faced in traditional tournament management methods. In manual systems, team registration, match scheduling, and result recording are often time-consuming and prone to errors. There is a lack of proper data organization, which can lead to confusion, duplication, and loss of information. Communication between organizers and participants is also inefficient, causing delays in updates and announcements. Additionally, maintaining accurate records and generating rankings manually is difficult and unreliable. This system addresses these issues by providing an automated, accurate, and user-friendly solution for efficient tournament management.

### 4.2. System Design

The Sports Tournament Team Registration and Match Result System is designed using a modular and structured approach to ensure efficiency and scalability. The system architecture is divided into different components such as user interface, application logic, and database management. The front-end interface allows users to interact with the system through forms and dashboards, while the backend processes handle business logic like registration, scheduling, and result calculation. A centralized database is used to store all tournament-related data securely, including team details, player information, match fixtures, and results. The design also includes input validation, error handling, and security measures to ensure data accuracy and system reliability. Overall, the system design supports smooth operation, easy maintenance, and effective tournament management.

### 4.3. Program Development

The Sports Tournament Team Registration and Match Result System is developed using a systematic approach to ensure efficiency and reliability. The development process includes planning, designing, coding, testing, and implementation. Programming languages such as Java are used to build the application, along with database technologies for data storage and management. Each module, including registration, scheduling, and result processing, is coded separately and then integrated to form the complete system. Proper testing is conducted to identify and fix errors, ensuring smooth performance. The program is designed to be user-friendly, secure, and capable of handling tournament operations effectively.

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### 4.4. Testing and Validation

The Sports Tournament Team Registration and Match Result System undergoes thorough testing and validation to ensure its accuracy, reliability, and performance. Different types of testing, such as unit testing, integration testing, and system testing, are carried out to verify that each module functions correctly. The system is tested with various inputs to check for errors, bugs, and unexpected behavior. Validation ensures that the system meets user requirements and produces correct outputs, such as accurate match schedules and results. Any identified issues are fixed during the testing phase to improve system quality. Overall, this process ensures that the application operates smoothly and efficiently in real-world scenarios

### 4.5. Result and Documentation

The Sports Tournament Team Registration and Match Result System successfully provides an efficient and reliable solution for managing sports tournaments. The system ensures accurate team registration, organized match scheduling, and quick result updates, improving overall performance and reducing manual work. The results demonstrate that the system is user-friendly, time-saving, and effective in handling tournament activities. Proper documentation is prepared to describe the system design, modules, functionality, and usage guidelines. This documentation helps users and developers understand the system easily and supports future maintenance, updates, and enhancements

**4.6. Algorithm**

## 1 Start the Program

**Display the main menu (Register Team, Schedule Match, Update Result, View Results).**

**2 If user selects Register Team:**

**Enter team and player details.**

**Store the details in the database.**

**3 If user selects Schedule Match:**

**Select registered teams.**

**Generate match fixtures.**

**Store match schedule.**

**4 If user selects Update Result:**

**Enter match score and result.**

**Update the database.**

**5 If user selects View Results:**

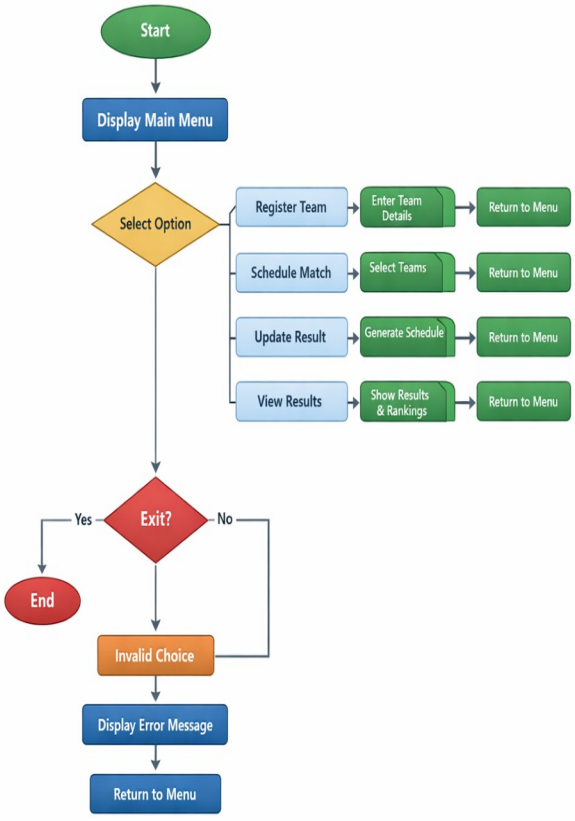
**Retrieve match results and rankings.**

**Display the information.**

**6 Repeat the process until the user exits.**

**7 Stop the program.**

## 4.7. Flowchart



# SOURCE CODE

```
import java.util.*;
class Team {
    String teamName;
    ArrayList<String> players;

    Team(String name) {
        teamName = name;
        players = new ArrayList<>();
    }
}

public class TournamentSystem {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Team> teams = new ArrayList<>();
    static HashMap<String, String> results = new HashMap<>();

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n--- Sports Tournament System ---");
            System.out.println("1. Register Team");
            System.out.println("2. View Teams");
            System.out.println("3. Schedule Match");
            System.out.println("4. Update Result");
            System.out.println("5. View Results");
            System.out.println("6. Exit");
```

```
        System.out.print("Enter choice: ");
        choice = sc.nextInt();
        sc.nextLine();

        switch (choice) {
            case 1:
                registerTeam();
                break;
            case 2:
                viewTeams();
                break;
            case 3:
                scheduleMatch();
                break;
            case 4:
                updateResult();
                break;
            case 5:
                viewResults();
                break;
            case 6:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid choice!");
        }
    } while (choice != 6);
}
```

```
static void registerTeam() {
    System.out.print("Enter Team Name: ");
    String name = sc.nextLine();
    Team team = new Team(name);

    System.out.print("Enter number of players: ");
    int n = sc.nextInt();
    sc.nextLine();

    for (int i = 0; i < n; i++) {
        System.out.print("Enter player " + (i + 1) + ": ");
        team.players.add(sc.nextLine());
    }

    teams.add(team);
    System.out.println("Team Registered Successfully!");
}

static void viewTeams() {
    if (teams.isEmpty()) {
        System.out.println("No teams registered.");
        return;
    }

    for (Team t : teams) {
        System.out.println("\nTeam: " + t.teamName);
        System.out.println("Players: " + t.players);
    }
}
```

```
static void scheduleMatch() {
    if (teams.size() < 2) {
        System.out.println("Not enough teams!");
        return;
    }

    System.out.println("Available Teams:");
    for (int i = 0; i < teams.size(); i++) {
        System.out.println(i + ". " + teams.get(i).teamName);
    }

    System.out.print("Select Team 1 index: ");
    int t1 = sc.nextInt();
    System.out.print("Select Team 2 index: ");
    int t2 = sc.nextInt();
    sc.nextLine();

    String match = teams.get(t1).teamName + " vs " + teams.get(t2).teamName;
    results.put(match, "Pending");

    System.out.println("Match Scheduled: " + match);
}

static void updateResult() {
    if (results.isEmpty()) {
        System.out.println("No matches found.");
        return;
    }
```

```
        for (String match : results.keySet()) {
            System.out.println(match);
        }

        System.out.print("Enter match name: ");
        String matchName = sc.nextLine();

        if (results.containsKey(matchName)) {
            System.out.print("Enter result: ");
            String res = sc.nextLine();
            results.put(matchName, res);
            System.out.println("Result Updated!");
        } else {
            System.out.println("Match not found!");
        }
    }

    static void viewResults() {
        if (results.isEmpty()) {
            System.out.println("No results available.");
            return;
        }

        for (Map.Entry<String, String> entry : results.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }
}
```

# INPUT AND OUTPUT SCREENS

* Main Menu Screen

========== SPORTS TOURNAMENT SYSTEM ==========

---------- MAIN MENU ----------

1. Register Team

2. View Teams

3. Schedule Match

4. Update Result

5. View Results

6. Exit

Enter your choice: 1

Enter Team Name: Chennai Kings

Enter number of players: 3

Enter player 1: Arun

Enter player 2: Karthik

Enter player 3: Rahul

Team Registered Successfully!

# 4. DISCUSSION & CONCLUSION

## 7.1 Discussion

The Sports Tournament Team Registration and Match Result System demonstrates how automation can significantly improve the management of sports events. Through the development of this system, it is observed that manual errors are reduced and the overall efficiency of tournament organization is increased. The system simplifies complex tasks such as team registration, match scheduling, and result processing. It also ensures better data management and quick access to information. However, the effectiveness of the system depends on proper data input and user interaction. The project highlights the importance of using technology in event management and provides a foundation for future improvements such as adding advanced features and expanding system capabilities.

## 7.2 Conclusion

The Sports Tournament Team Registration and Match Result System successfully achieves its goal of simplifying and automating tournament management. The system provides an efficient way to handle team registration, match scheduling, and result updates with accuracy and ease. It reduces manual effort, minimizes errors, and improves the overall organization of sports events. The user-friendly design makes it accessible for both administrators and participants. This project demonstrates the importance of digital solutions in managing real-world activities and can be further enhanced with additional features for larger and more complex tournaments

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